

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

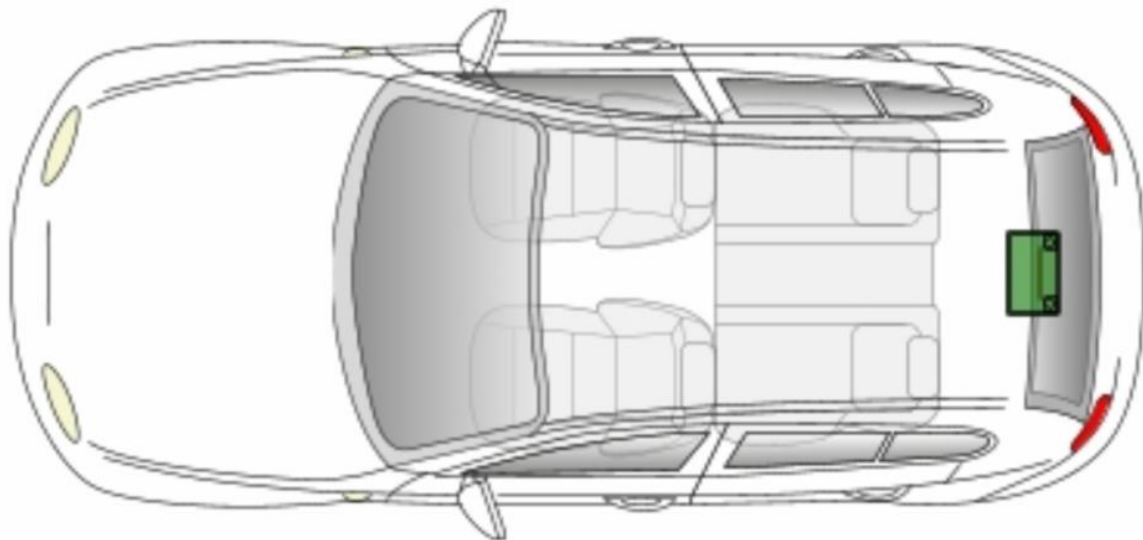
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

Battery location



Luggage compartment

Clean off the old sealant

Renew the sealant

Removal

Remove the engine

Disconnect the pressure lines

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Remove the fuel lines from the high-pressure fuel pump

Remove the fuel rail

Remove the injectors

Remove the cylinder head cover

Remove the flywheel

Remove the oil sump

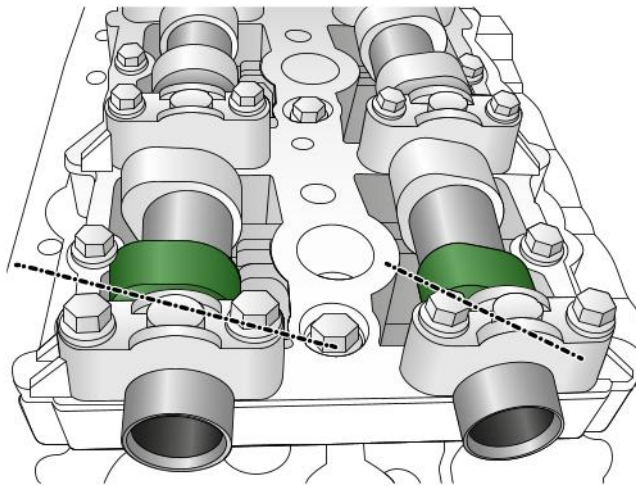
Disconnect the crankshaft position sensor

Remove the timing chain cover

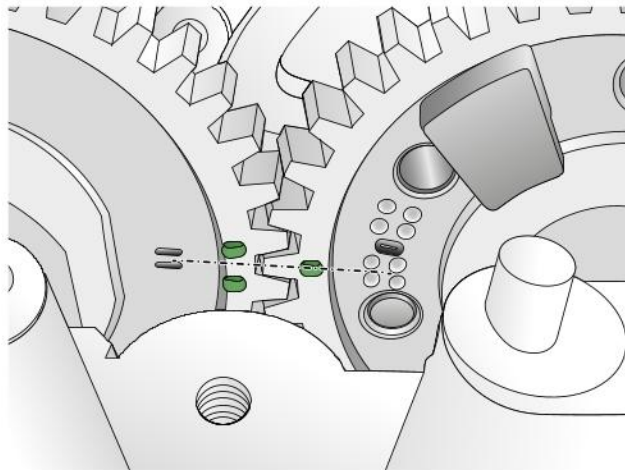
Turn the crankshaft to TDC for cylinder 1

Note: Only reposition the crankshaft to TDC by rotating it clockwise

Note: Both camshafts must point upwards at an angle

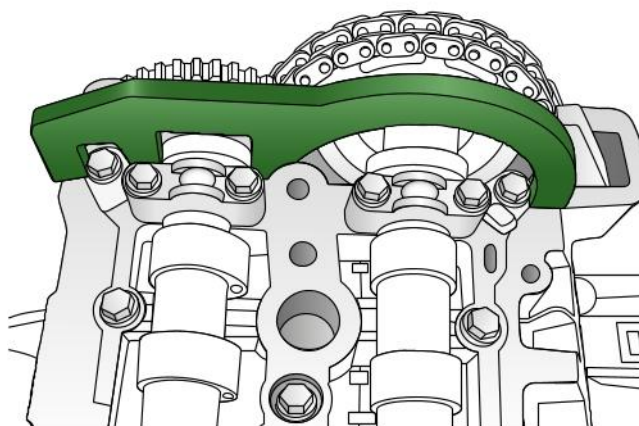


Check the camshaft timing marks

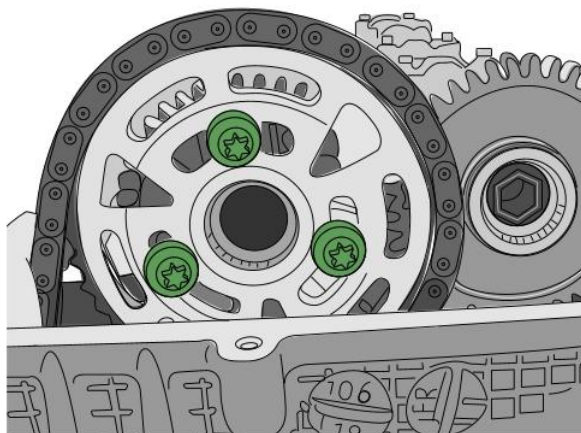


Fit the camshaft locking tool (11 8 760)

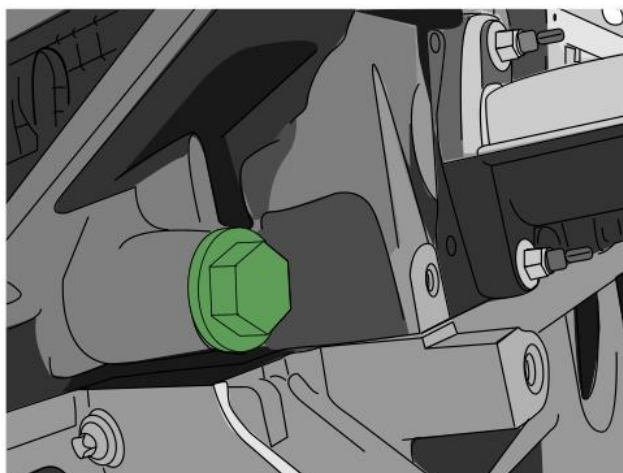
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Remove the camshaft gearwheel bolts



Remove the chain tensioner

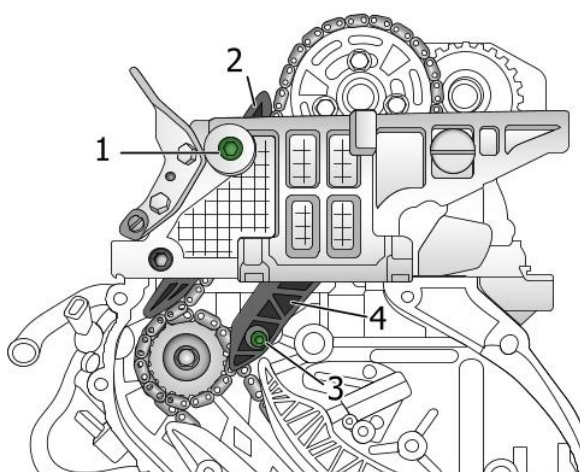


Remove the chain guide

Remove the chain tensioner guide

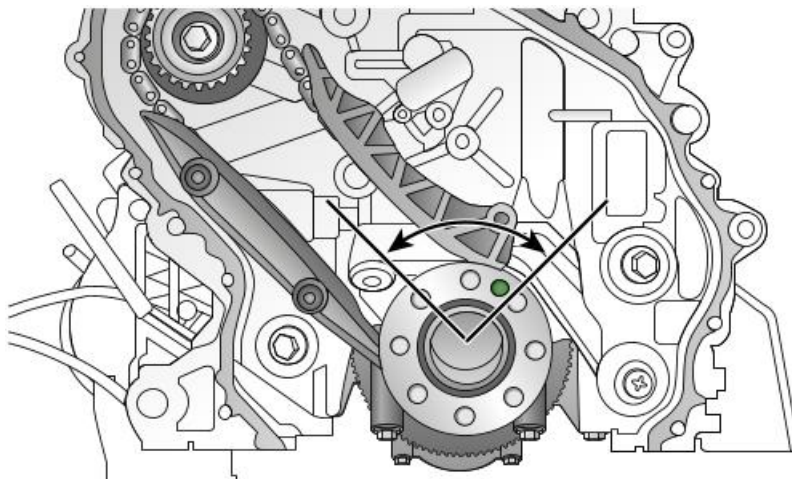
Removal sequence: (1-3-2-4)

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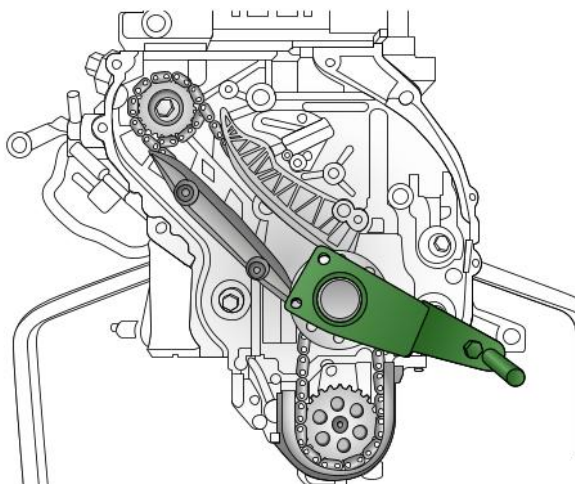
Remove the timing chain together with the camshaft gearwheel

Note: Keep the guide pin in the area shown during repair



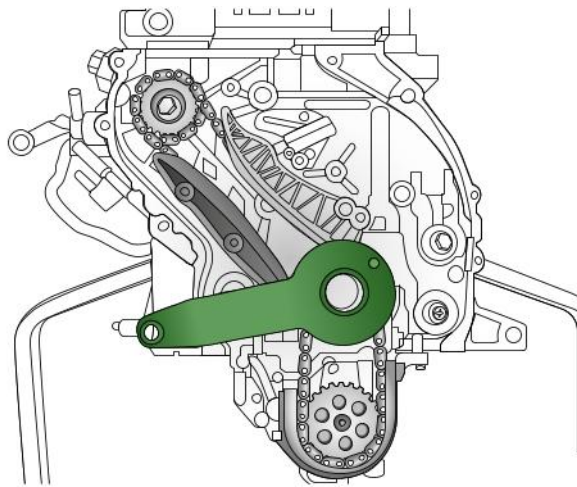
Vehicles up to: (5/2013)

Fit the crankshaft locking tool (11 4 320)



Vehicles from: (6/2013)

Fit the crankshaft locking tool (2 285 671)

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All vehicles:

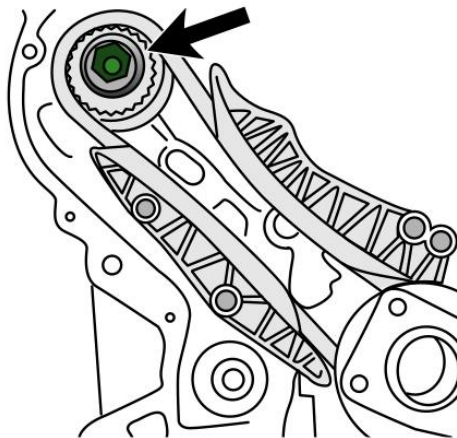
Loosen the oil pump gearwheel

Note: Left-hand thread

Loosen the high-pressure pump gearwheel bolt

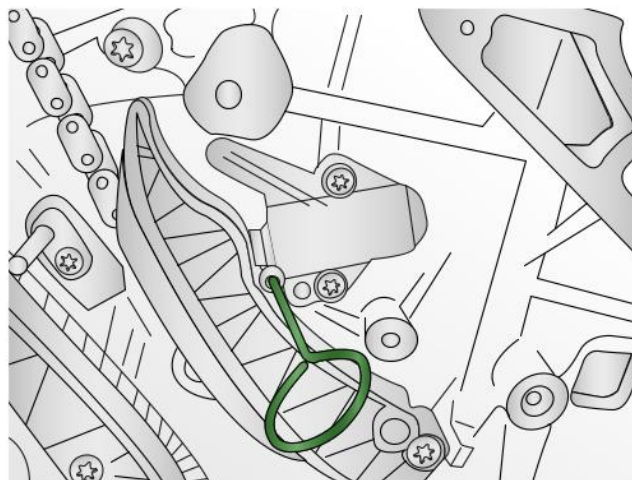
Fit the special tool (2 448 914)

Detach the high-pressure pump gearwheel



Compress the secondary chain tensioner

Lock the chain tensioner with a locking pin (11 4 120)

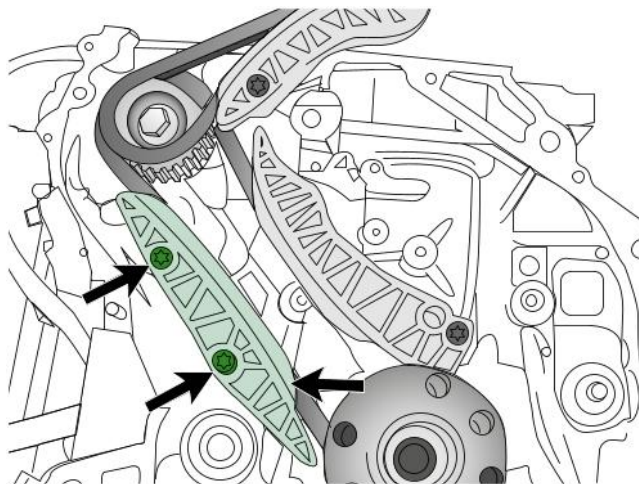


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Remove the secondary chain tensioner

Remove the chain guide

Remove the chain tensioner guide



Remove the crankshaft locking tool

Remove the oil pump gearwheel

Remove the oil pump chain

Remove the secondary timing chain

Installation

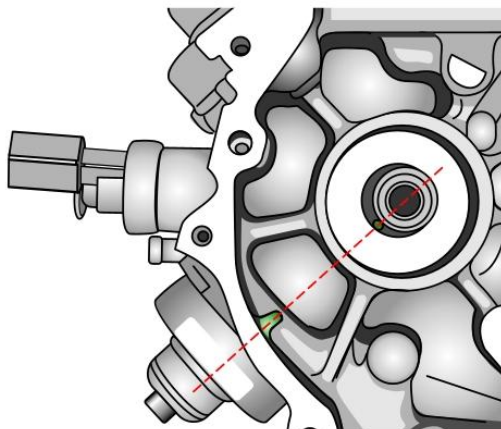
Secondary timing chain:

Refit the secondary timing chain

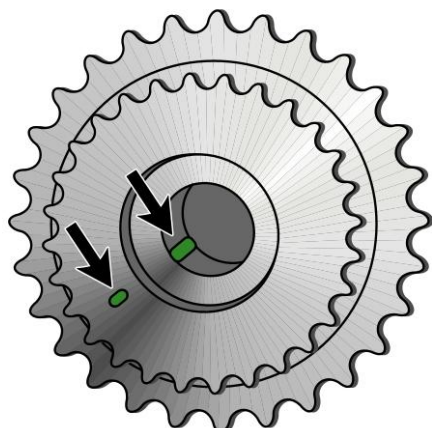
Fit the oil pump chain onto the crankshaft gearwheel

Check the position of the special tool (11 4 320, 2 285 671)

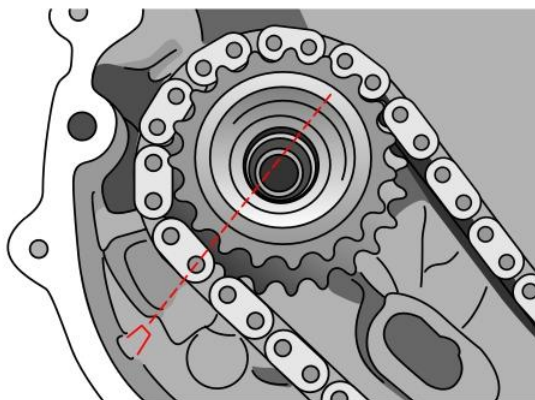
Check the timing marks



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Refit the high-pressure pump gearwheel



Note: Do not tighten the bolt

Refit the chain guide

Refit the chain tensioner guide

Refit the bearing pins for the chain guide rail and the tensioner rail

Refit the secondary timing chain tensioner

Refit the oil pump gearwheel

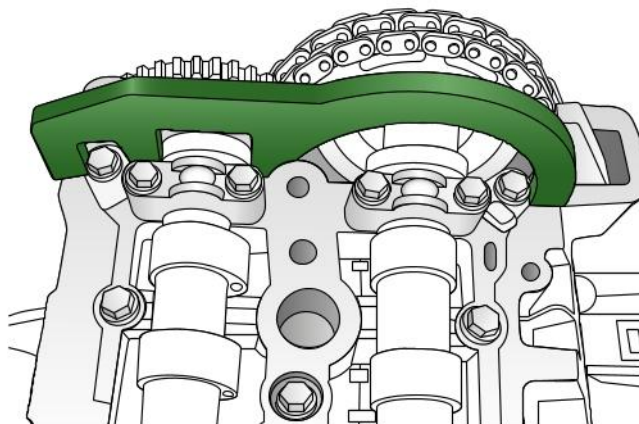
Fit the oil pump chain onto the oil pump gearwheel

Tighten the high-pressure pump gearwheel

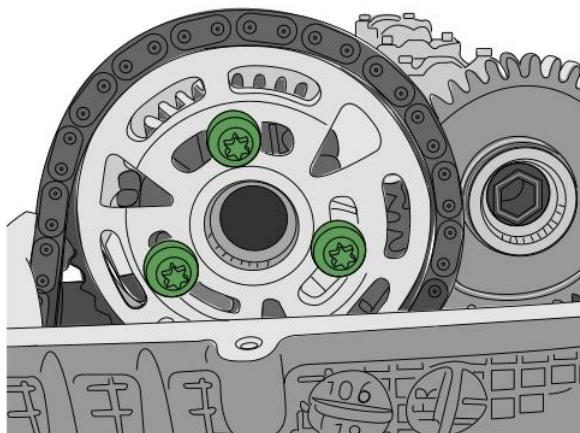
Primary timing chain:

Check that the camshaft locking tool is in position (11 8 760)

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Fit the camshaft gearwheel and chain



Note the position of the bolts

Refit the chain guide

Refit the chain tensioner guide

Refit the bearing pins for the chain guide rail and the tensioner rail

Tighten the camshaft gearwheel bolts (10 Nm)

Loosen the camshaft gearwheel bolts (90°)

Refit the chain tensioner

Remove the crankshaft locking tool (11 4 320, 2 285 671)

Timing chain cover:

Apply sealant (Loctite 5970)

Note: The cover must be fitted within 10 minutes of applying the sealant

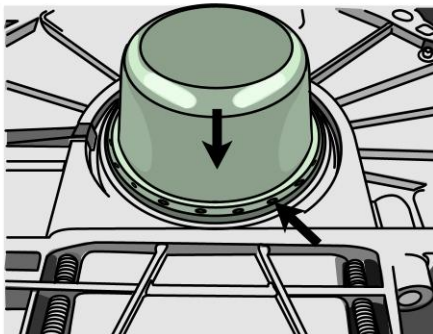
Refit the timing cover

Crankshaft oil seal:

Use the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Fit the special tool: (11 8 815)

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Note: Do not apply grease

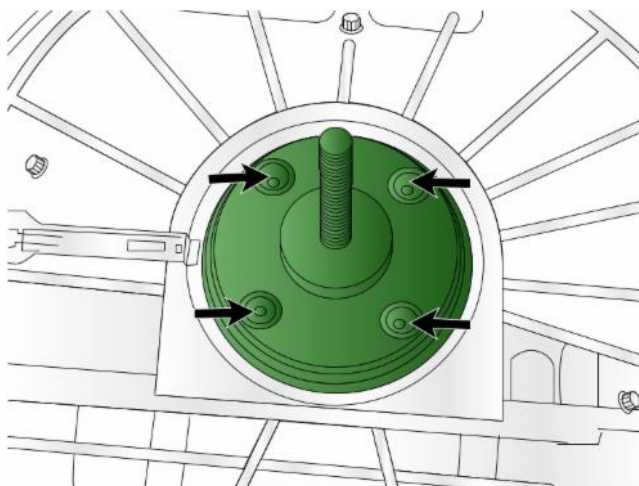
Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

Remove the special tool (11 8 815)

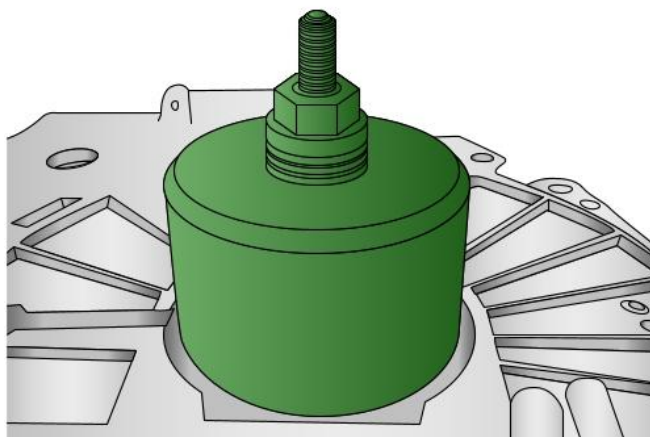
Fit the special tool: (11 8 811)

Tighten the bolts (11 8 812)



Fit the special tool: (11 8 813)

Tighten the nut (11 8 814)



Note: Until the crankshaft seal is level with the timing cover

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Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Refit the flywheel

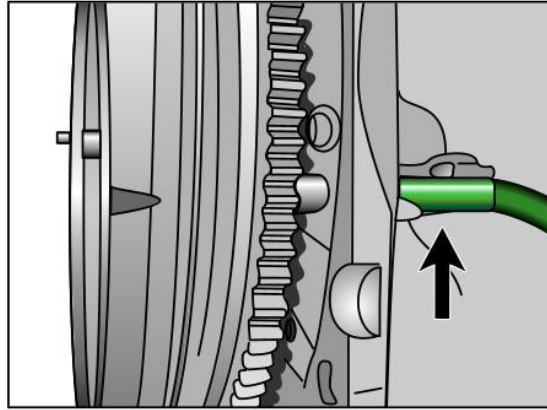
Fit the flywheel locking tool (11 9 260)

Tighten the flywheel bolts

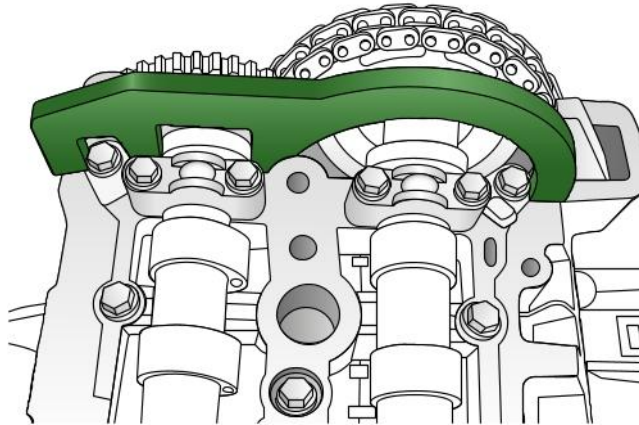
Remove the flywheel locking tool (11 9 260)

Turn the crankshaft to TDC for cylinder 1

Lock the crankshaft by fitting a locking pin into the flywheel (11 5 320)



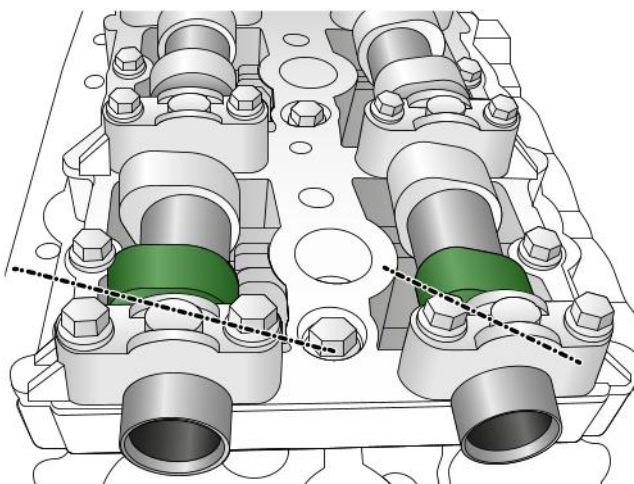
Check the position of the special tool



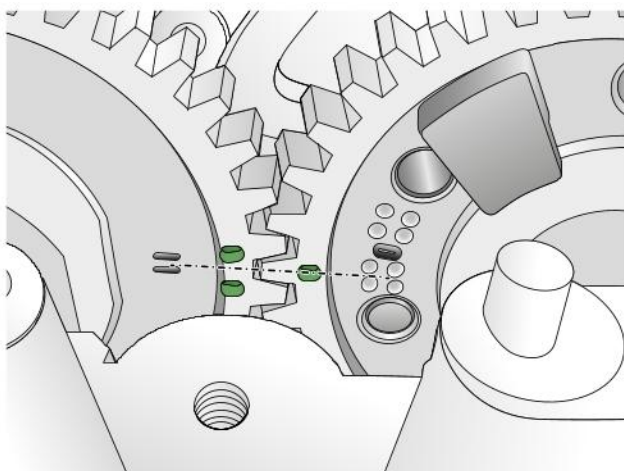
Tighten the camshaft gearwheel bolts

Note: Both camshafts must point upwards at an angle

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Check the camshaft timing marks



Remove all the locking tools

Turn the engine 2 rotations clockwise

Check the timing again

Further refitting is in the reverse order of removal

Refit the engine

Torque settings

Starter motor

Starter motor 19 (Nm)

Fuel rail

Fuel rail 24 (Nm)

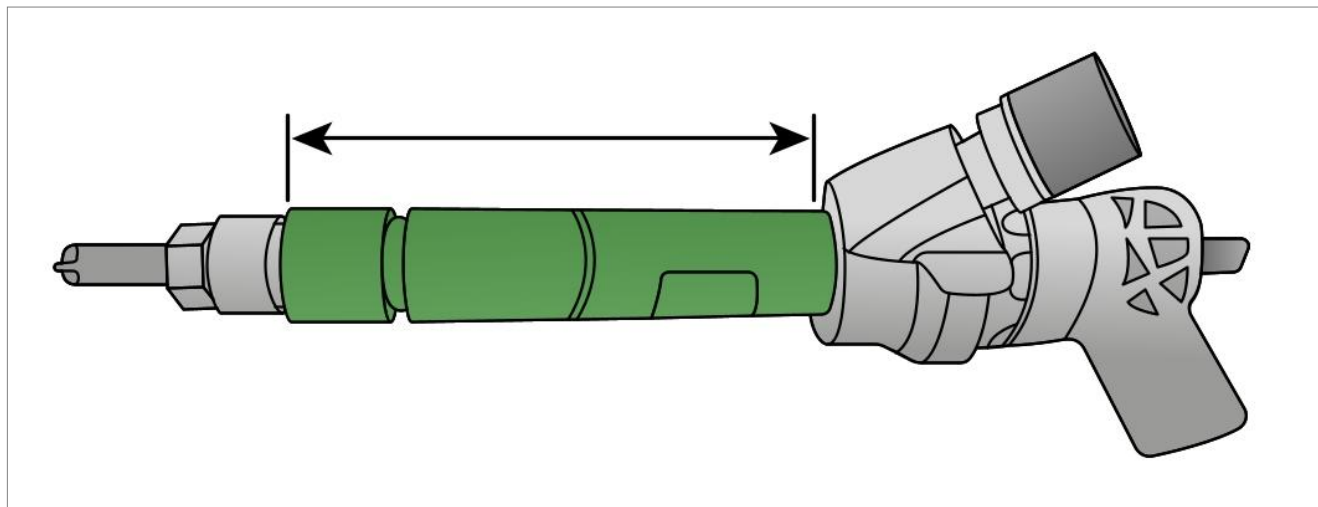
Injector

Injector (Renew the sealing washer)

Clean the contact surfaces

Apply high-temperature paste to the area shown

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Stage 1 8 (Nm)

Stage 2 26 (Nm)

Valve cover

Valve cover 10 (Nm)

Flywheel

Oil sump

Sump 24 (Nm)

Engine oil drain plug

Engine oil drain plug (Renew the sealing washer) 25 (Nm)

High-pressure pump gearwheel

High-pressure pump gearwheel 65 (Nm)

Oil pump gearwheel

Oil pump gearwheel (Renew the bolt (left-hand thread))

Stage 1 5 (Nm)

Stage 2 90 (°)

Timing chain tensioner

Timing chain tensioner (Upper) 70 (Nm)

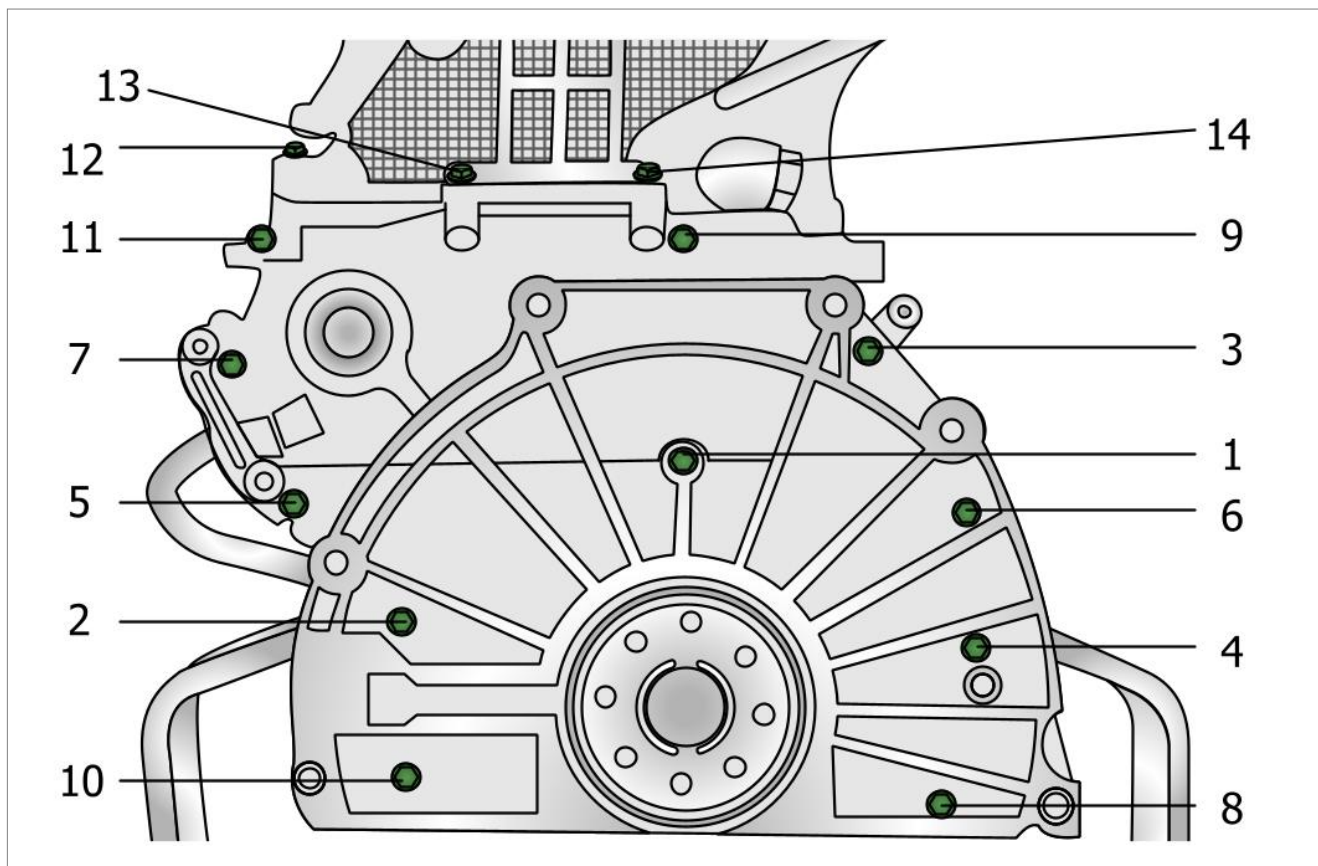
Timing chain tensioner (Lower) 10 (Nm)

Timing chain tensioner guide

Timing chain tensioner guide 20 (Nm)

Timing cover

Timing cover (Renew the bolts)

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Timing cover (Bolts 1 - 11)

Stage 1 8 (Nm)

Stage 2 90 (°)

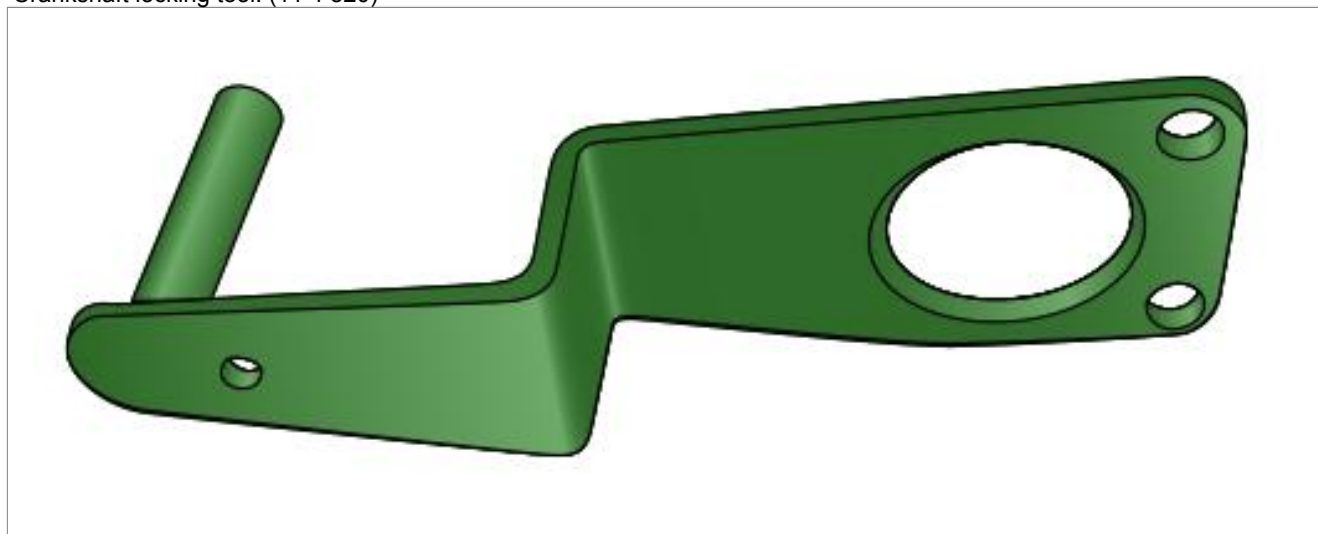
Timing cover (Bolts 12 - 14) 19 (Nm)

Camshaft gearwheel

Camshaft gearwheel 17 (Nm)

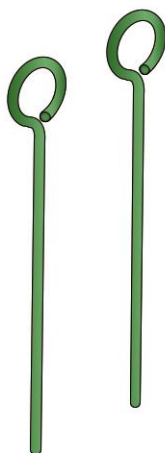
Special tools

Crankshaft locking tool: (11 4 320)

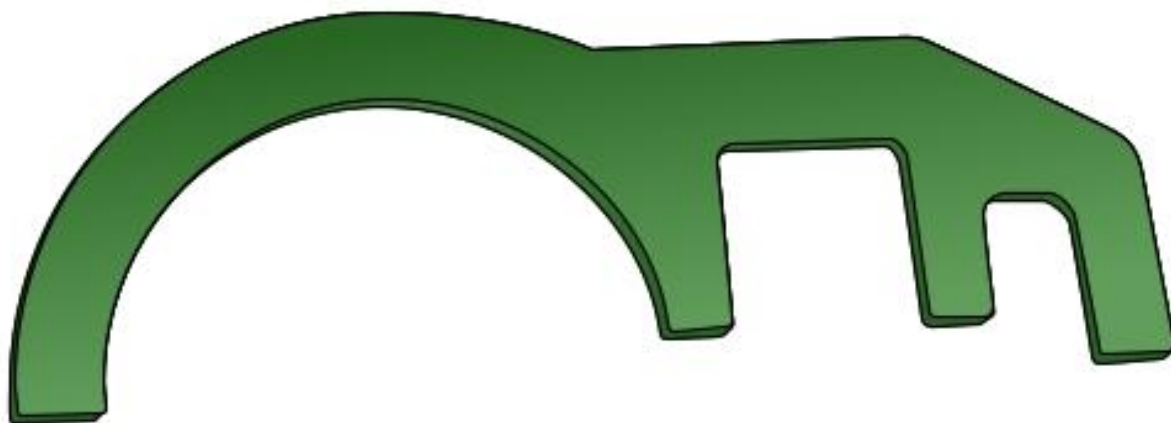


Locking pin: (11 4 120)

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Camshaft blocking tool: (11 8 760)

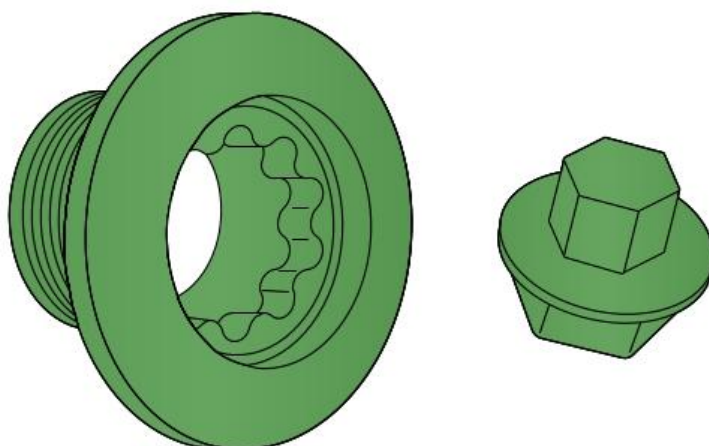


Crankshaft locking tool: (2 285 671)

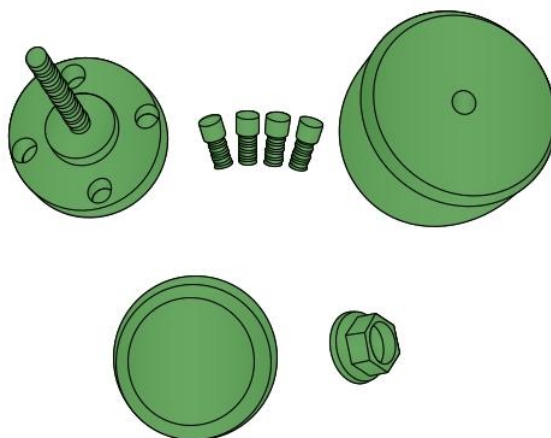


Extractor: (2 448 914)

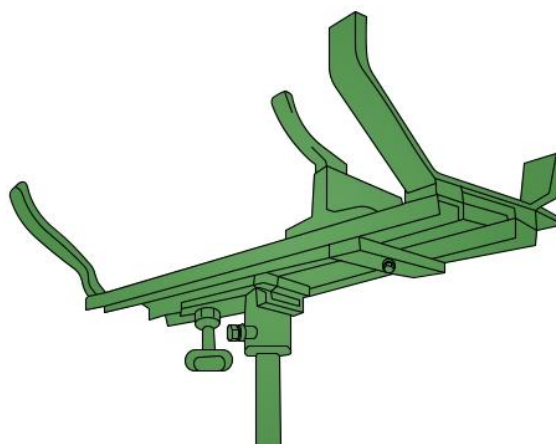
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Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

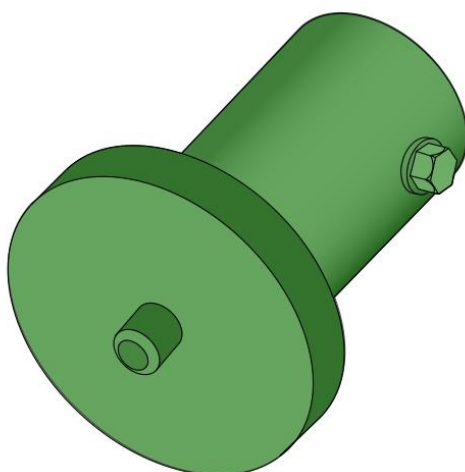


Gearbox support: (23 4 050, 00 2 030)

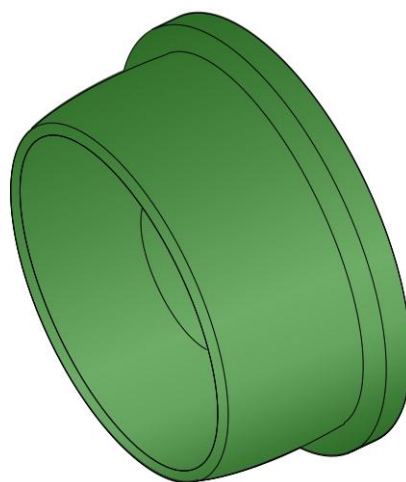


Support (31 2 220)

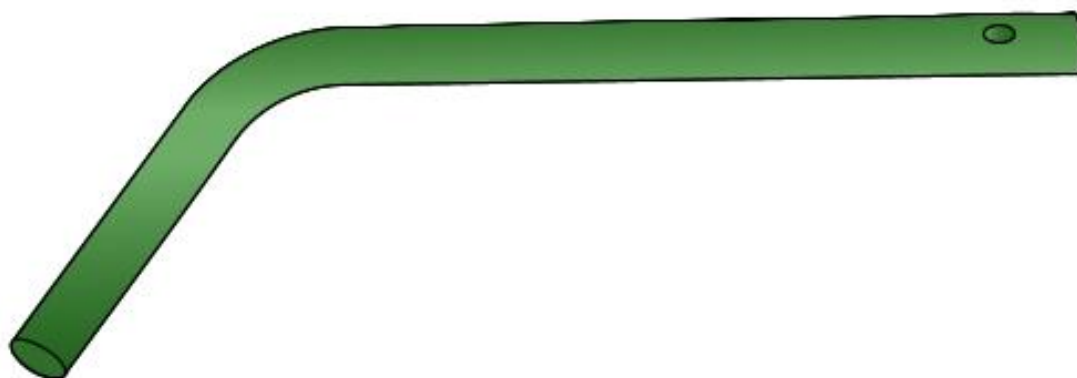
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Plug (24 3 390)

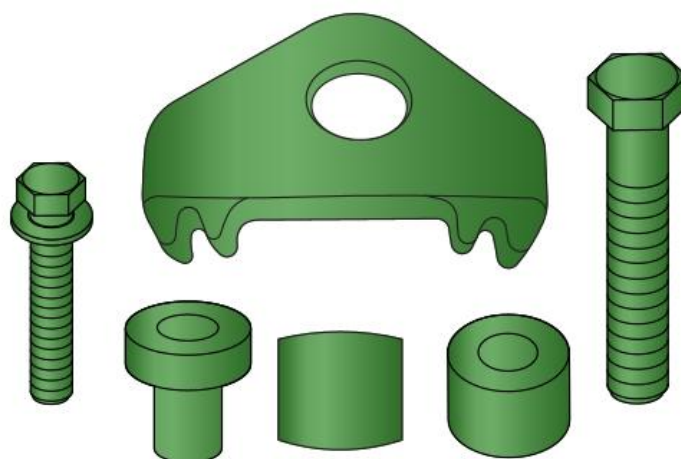


Flywheel locking pin: (11 5 320)



Flywheel locking tool: (11 9 260)

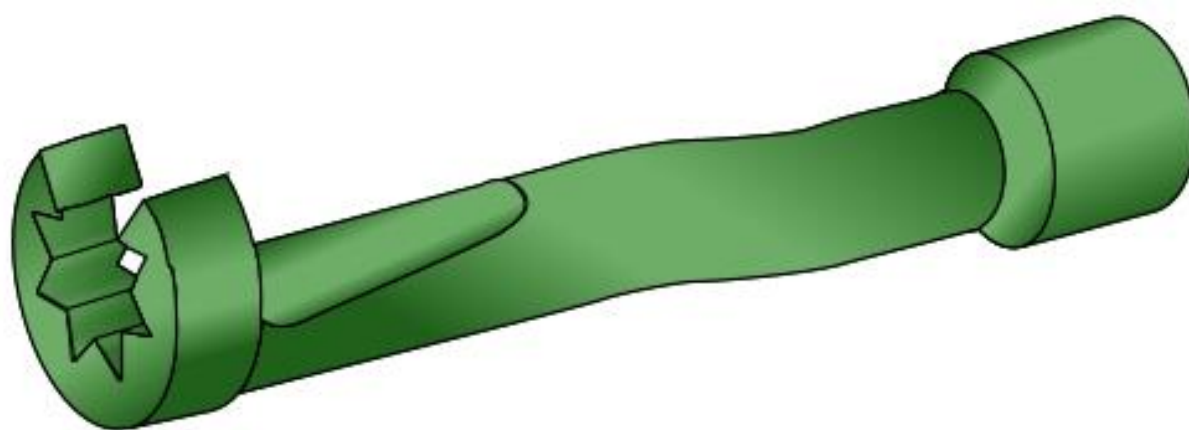
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Impact extractor: (13 6 100)



Socket wrench: (37 1 152)



Injector cleaner: (13 5 215)

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Injector cleaner: (13 5 210)

